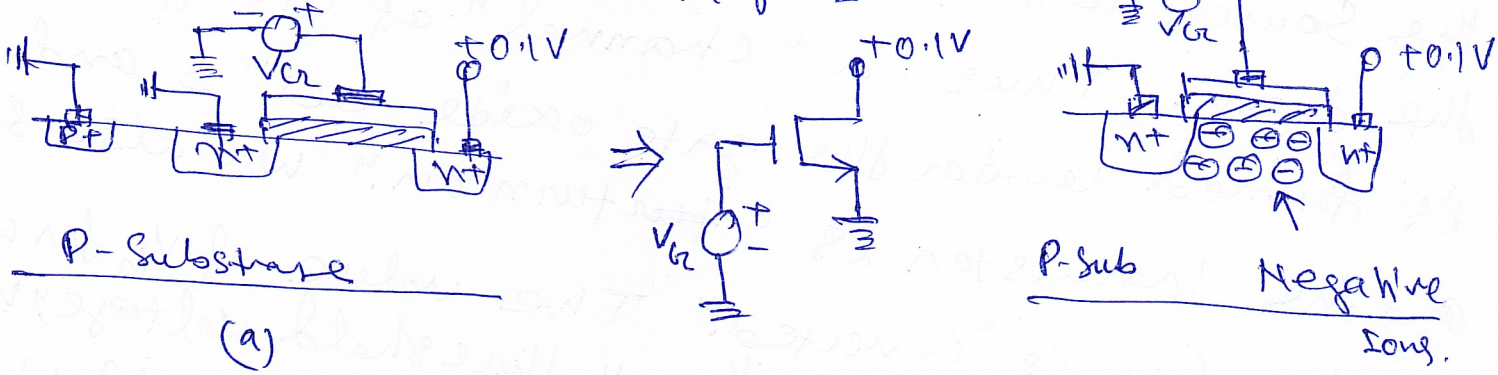
MOS I/V characteristics:

We analyze the generation and transport of charge in MOSFETs as a function of the terminal voltage. Our objective is to derive equations for the I/V characteristics such as abstraction from device level to circuit level.

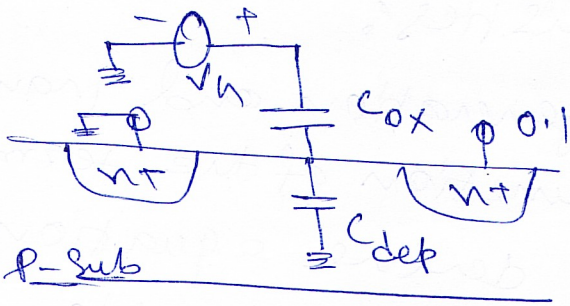
1. Threshold voltage:

Consider an NFET connected to external voltages as shown in Fig. 2 a.



What happens as the gate voltage, V_G, increases from zero? Since the gate and the substrate form a capacitor, as V_G becomes more positive, the holes in the p-substrate repelled from the gate area, leaving negative ions behind so as to minor the charge on the gate. In other words, a depletion region is created Fig. b. Under this condition, no current flows because no charge carrier are available.

As V_G increases, so to the width of the depletion region and the potential at the oxide silicon interface



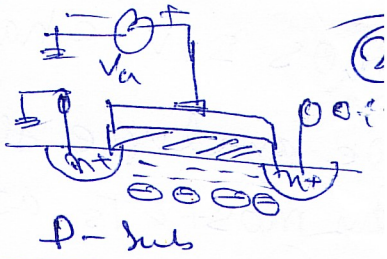


Fig. Formation of inversion layer

In a sense, the structure resembles two capacitors in series: the gate oxide capacitor and the depletion region capacitor. When the interface potential reaches a sufficiently positive value, electrons flow from the source to the interface and eventually to the drain. Thus, a "channel" of charge carriers is formed under the gate oxide between S and D and the transistor is "turn on." We also say the interface is "inverted." The value of V_G for which this occurs is called the "threshold voltage," V_{TH}. If V_G rises further, the charge in the depletion region remains relatively constant while the channel charge density continues to increase, providing a greater current from S to D.

In reality, the turn-on phenomenon is a gradual function of the gate voltage, making it difficult to define V_{TH} unambiguously. In semiconductor physics, the V_{TH} of an NFET is usually defined as the gate voltage for which the

the interface is "as much n-type as the sub is p-type" it can be proved that

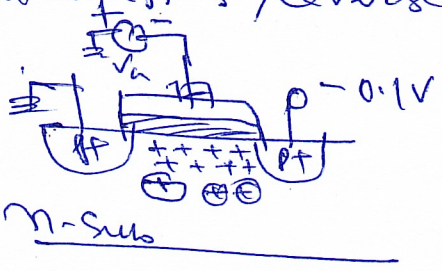
$$V_{TH} = \Phi_{MS} + 2\Phi_F + \frac{Q_{dep}}{C_{ox}} \quad \cdots (1)$$

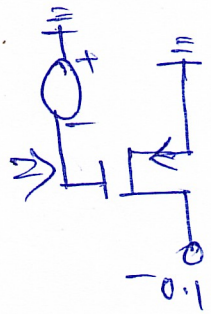
where Φ_{MS} is the difference betn the work functions of the polysilicon gate and the Silicon Substrate, $\Phi_F = \left(\frac{kT}{q}\right) \ln\left(N_{sub}/n_i\right)$,

q is electron charge, N_{sub} is the doping concentration of the substrate, Q_{dep} is the charge in the depletion region, and C_{ox} is the gate oxide capacitance per unit area.

In our preliminary analysis that the device turns on abruptly for $V_{GS} \geq V_{TH}$.

On turn-on phenomenon in PMOS device is similar to that of MIFETs but with all of the polarities reversed.





of the gate-source voltage becomes sufficiently negative, an inversion layer consisting of holes is formed at the oxide silicon interface, providing a conduction path betn the source and the drain.